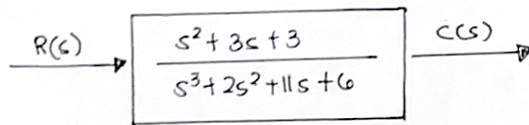


3.)



$$\frac{C(s)}{R(s)} = \frac{s^2 + 3s + 3}{s^3 + 2s^2 + 11s + 6}$$

$$C(s) [s^3 + 2s^2 + 11s + 6] = R(s) [s^2 + 3s + 3]$$

$$\mathcal{L}^{-1} \{ s^3 C(s) + 2s^2 C(s) + 11s C(s) + 6C(s) = s^2 R(s) + 3s R(s) + 3R(s) \}$$

$$\ddot{c} + 2\ddot{c} + 11\dot{c} + 6c = \ddot{r} + 3\dot{r} + 3r$$

$$x_1 = c(t) \quad \dot{x}_1 = \dot{c}(t) = x_2 \quad u_1 = r$$

$$x_2 = \dot{c}(t) \quad \dot{x}_2 = \ddot{c}(t) = x_3$$

$$x_3 = \ddot{c}(t) \quad \dot{x}_3 = \dddot{c}(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u_1$$